

Control work 5 — angles

Angles on a line, at a point, in triangles and in quadrilaterals.

LESSON 87

1 × 40 MIN

MATEMATIKA 6, NAZORAT ISHI 5

S7 5 REVIEW

LESSON CLOCK

2'

28'

8'

2'

Instructions	2 min
The paper	28 min
Answers and diagnosis	8 min
What comes next	2 min

LEARNING OBJECTIVES

- Classify and measure angles accurately.
- Use the four angle facts with a reason each time.
- Find angles in triangles and quadrilaterals.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 223–246
Cambridge	Stage 7, pp. 55–62
Workbook	Control paper 5

01

Explanation

The paper — 20 marks, 28 minutes

Q	TASK	MARKS	FROM
1	Classify 37° , 90° , 128° and 245°	2	L80
2	x and 118° on a straight line: find x with a reason	3	L81–83
3	x , $3x$ and 140° at a point: find x	3	L81–83
4	$5x$ and $2x + 60$ vertically opposite: find the angle	3	L81–83
5	A triangle with 72° and 43° : find the third	3	L84–86
6	An isosceles triangle with apex 34° : find the base angles	3	L84–86
7	A quadrilateral with 85° , 95° and 110° : find the fourth	3	L84–86

THE ANSWERS

acute, right, obtuse, reflex; 62° ; 55° ; 100° ; 65° ; 73° each; 70° .

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
reflex called obtuse	245°	over 180° is reflex
no reason given	$x = 62^\circ$	“angles on a straight line”
angles at a point summed to 180°	$x + 3x + 140 = 180$	$x + 3x + 140 = 360$
vertically opposite set equal to 180°	$5x + 2x + 60 = 180$	$5x = 2x + 60$
angle sum taken as 360° for a triangle	$360 - 115$	$180 - 115$
apex used as a base angle	$34^\circ, 34^\circ, 112^\circ$	$34^\circ, 73^\circ, 73^\circ$
quadrilateral summed to 180°	a negative answer	360°

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

What comes next

IF YOU LOST MARKS ON	IT RETURNS IN
Q1	every diagram for the rest of the year
Q2–Q4	the circle, where angles at the centre fill 360°
Q5–Q6	Grade 7, where these facts are proved
Q7	the pie chart in Quarter IV

LOOKING FORWARD

The rest of Quarter II is the circle: its parts, and the length of the curve round it. The 360° of this paper is what makes a pie chart possible.

02 Worked examples

EXAMPLE 1 Model answer, Q3: x , $3x$ and 140° at a point.

① $x + 3x + 140 = 360$

Angles at a point.

② $4x = 220$

③ $x = 55^\circ$

Check: $55 + 165 + 140 = 360$ ✓

ANSWER

55°

EXAMPLE 2 Model answer, Q4: $5x$ and $2x + 60$ vertically opposite.

① Vertically opposite angles are equal.

② $5x = 2x + 60$

③ $3x = 60$, so $x = 20$.

④ Each angle is 100° .

ANSWER

100°

EXAMPLE 3 Model answer, Q6: isosceles with apex 34° .

① $180 - 34 = 146$ for the two base angles.

② $146 \div 2 = 73$.

③ 73° each.

Check: $34 + 73 + 73 = 180$ ✓

ANSWER

73° each

03 Interactive model

Mark Q2 for the reason alone; the class discovers that the number without a reason is worth less than half the marks.

The chapter in eight questions

245° is:

acute

obtuse

straight

reflex

x with 118° on a line:

62°

72°

242°

118°

Its reason:

angles at a point

angles on a straight line

vertically opposite

triangle sum

$$x + 3x + 140 = \dots$$

So x is:

$5x$ and $2x + 60$ vertically opposite give the angle:

A triangle with 72° and 43° :

A quadrilateral with 85° , 95° , 110° :

60°

70°

80°

90°

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Acute angle	O‘tkir burchak	Острый угол
Obtuse angle	O‘tmas burchak	Тупой угол
Vertically opposite	Vertikal burchaklar	Вертикальные углы
Angle sum	Burchaklar yig‘indisi	Сумма углов
Isosceles	Teng yonli	Равнобедренный
Reason	Asos	Обоснование
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 tests:

calculating

classifying

measuring

drawing

Q2 needs, besides the number:

a diagram

a reason

a measurement

nothing

Q3 sums to:

 90° 180° 360° 540°

Q4 sets the two expressions:

equal

to 180°

to 360°

to 90°

Q6 divides by:

2

3

4

nothing

Q7 sums to:

180°

270°

360°

540°

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Classify 37°

ANSWER Acute

2. Classify 128°

ANSWER Obtuse

3. Classify 245°

ANSWER Reflex

4. x with 118° on a line

ANSWER 62°

5. x , $3x$, 140° at a point

ANSWER $x = 55^\circ$

6. A triangle with 72° and 43°

ANSWER 65°

7. A quadrilateral with 85° , 95° , 110°

ANSWER 70°

Medium · the standard question

7 PROBLEMS

1. $5x$ and $2x + 60$ vertically opposite

ANSWER 100°

2. Isosceles with apex 34°

ANSWER 73° each

3. Isosceles with base angle 34°

ANSWER Apex 112°

4. Two supplementary angles in the ratio 3 : 7

ANSWER 54° and 126°

5. An angle twice its complement

ANSWER 60°

6. A parallelogram with one angle 115°

ANSWER $115^\circ, 65^\circ, 115^\circ, 65^\circ$

7. The reflex angle beside 118°

ANSWER 242°

Hard · stretch and reason

7 PROBLEMS

1. A triangle with angles x , $x + 20$, $x + 40$

ANSWER $40^\circ, 60^\circ, 80^\circ$

2. A quadrilateral with angles x , $2x$, $3x$, $4x$

ANSWER $36^\circ, 72^\circ, 108^\circ, 144^\circ$

3. Four angles at a point, three equal and one 120°

ANSWER 80° each

4. An isosceles triangle with one angle of 90°

ANSWER $90^\circ, 45^\circ, 45^\circ$

5. A triangle cannot have two obtuse angles: why?

ANSWER They already exceed 180°

6. An exterior angle of a triangle beside 65°

ANSWER 115°

7. Why is a reason needed for every angle answer?

ANSWER The diagram is not evidence; the fact is

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on, with its reason.

1. Rewrite in full every question on which you lost a mark.
2. Find x if x and 143° lie on a straight line, giving a reason.
3. Find x if $2x$, $4x$ and 120° meet at a point.
4. An isosceles triangle has an apex of 50° . Find the base angles.
5. A quadrilateral has angles 100° , 75° and 95° . Find the fourth.